

GRIN3D mutation-hotspot prototype evaluation

TP53, SUZ12, and EZH2 | null-trial and AlphaFold pLDDT sensitivity

Question

Does the mutation prototype recover stable clusters in a dense positive-control protein and in proteins with sparser mutation patterns when null-simulation trials and structural-confidence filtering are varied?

Defensible conclusion

Observed cluster membership is deterministic and fully stable across 100-1,000 null trials at a fixed pLDDT threshold. Empirical p-values and significance calls are not uniformly converged by 1,000 trials. pLDDT filtering has a larger effect because it removes structural coordinates and mutation events, producing a different analyzed cohort. The prototype is suitable for cluster generation and exploratory sensitivity analysis, but final inference requires more null trials and independent-seed checks.

- 90 completed configurations; trial counts: 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000
- pLDDT thresholds: 0, 70, and 90 for TP53, SUZ12, and EZH2
- 25 of 25 automated validation checks passed
- Reference for each condition: its 1,000-trial result; membership reference across pLDDT: unfiltered 1,000-trial result

Study design and comparison logic

Protein	Events	Subjects	Protein residues	pLDDT available
TP53	30	23	393	Yes
SUZ12	72	57	739	Yes
EZH2	95	74	746	Yes

Varied settings

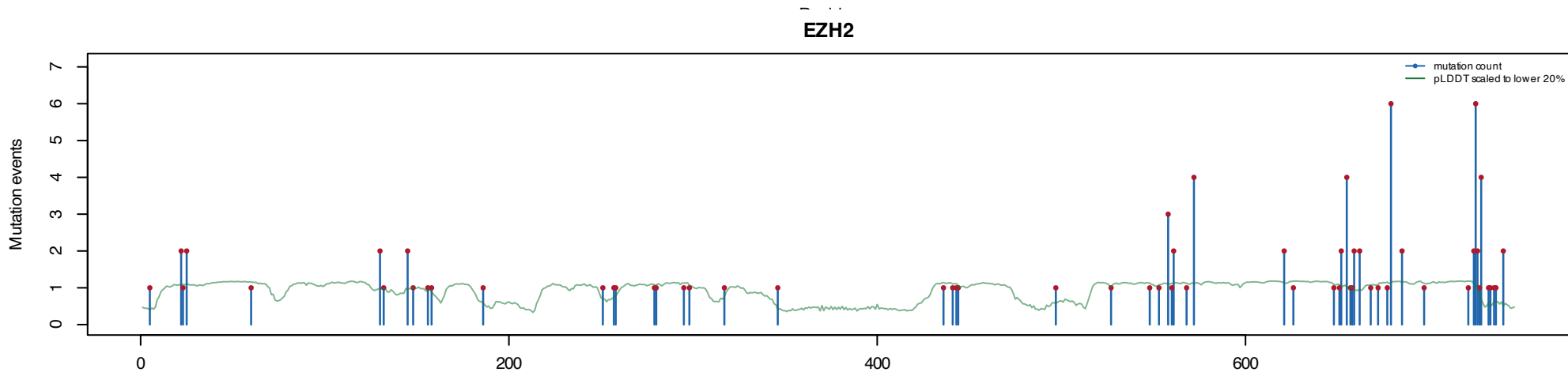
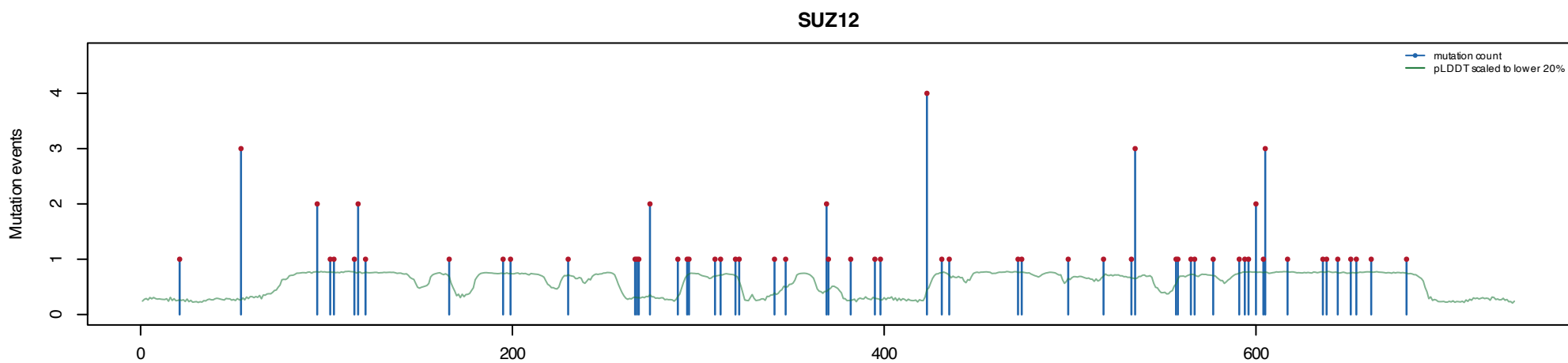
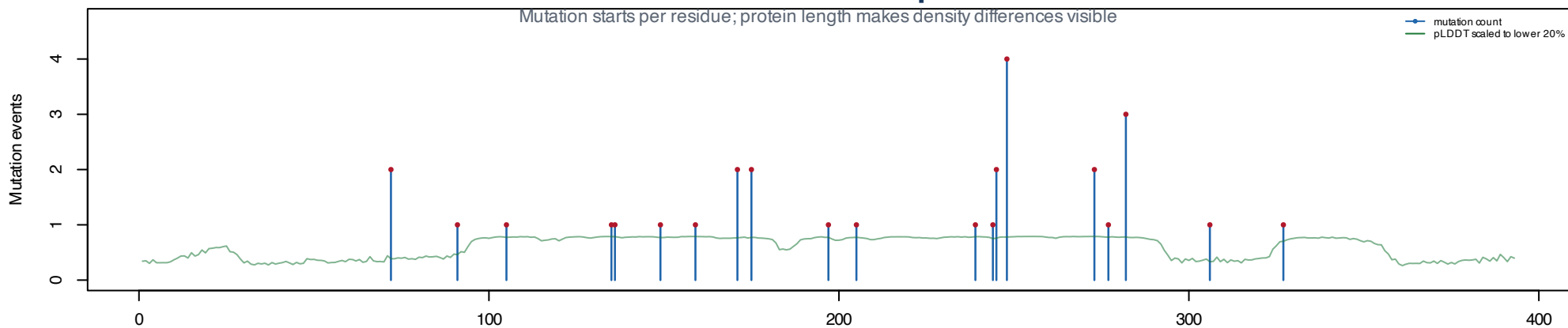
- Null simulations: 100 through 1,000 in increments of 100.
- Minimum pLDDT: 0 (unfiltered), 70 (confident), and 90 (very high confidence)

Held fixed

- Minimum support: 2 subjects and 2 events.
- Random seed 20260916; 50/50 calibration/evaluation split.
- Complete event mapping, within-subject overlap avoidance, and uniform positional null.

Interpretation boundary: trial count changes the simulated null and empirical p-values, not the observed clustering tree. pLDDT changes the eligible structure and may exclude mutation events; pLDDT comparisons therefore do not contain identical mutation cohorts.

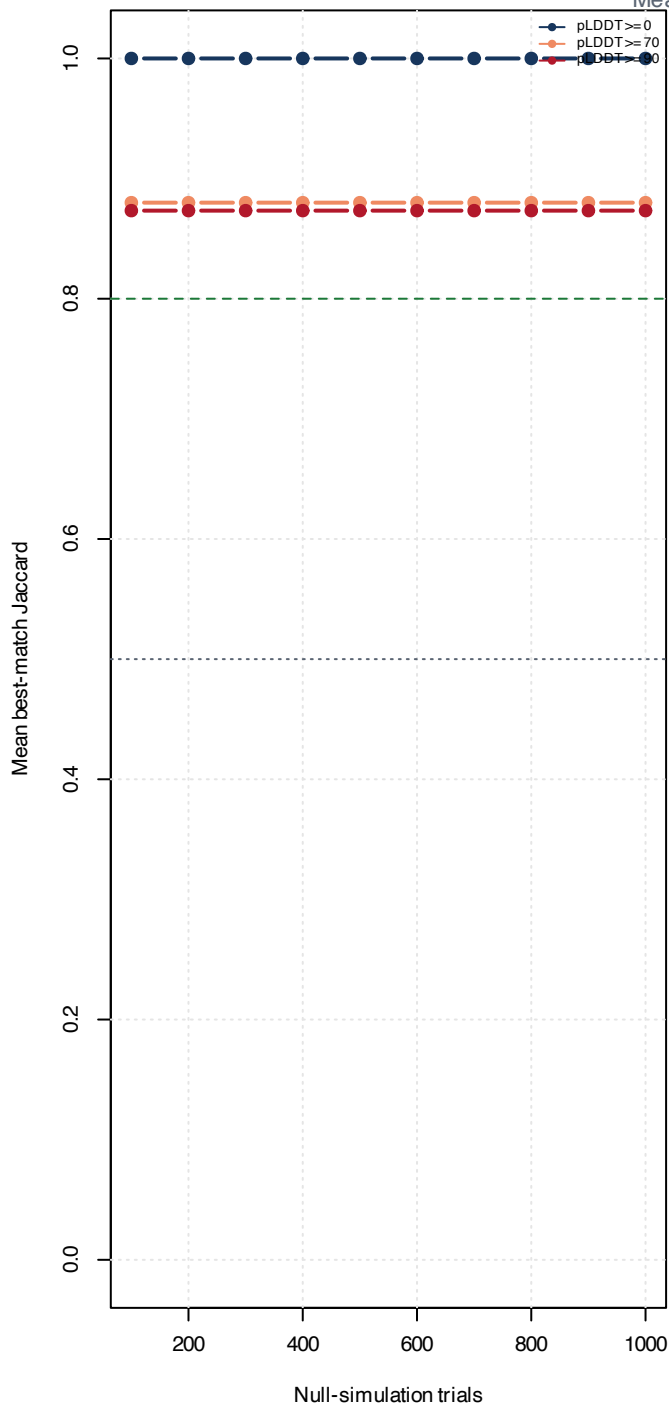
Mutation landscapes



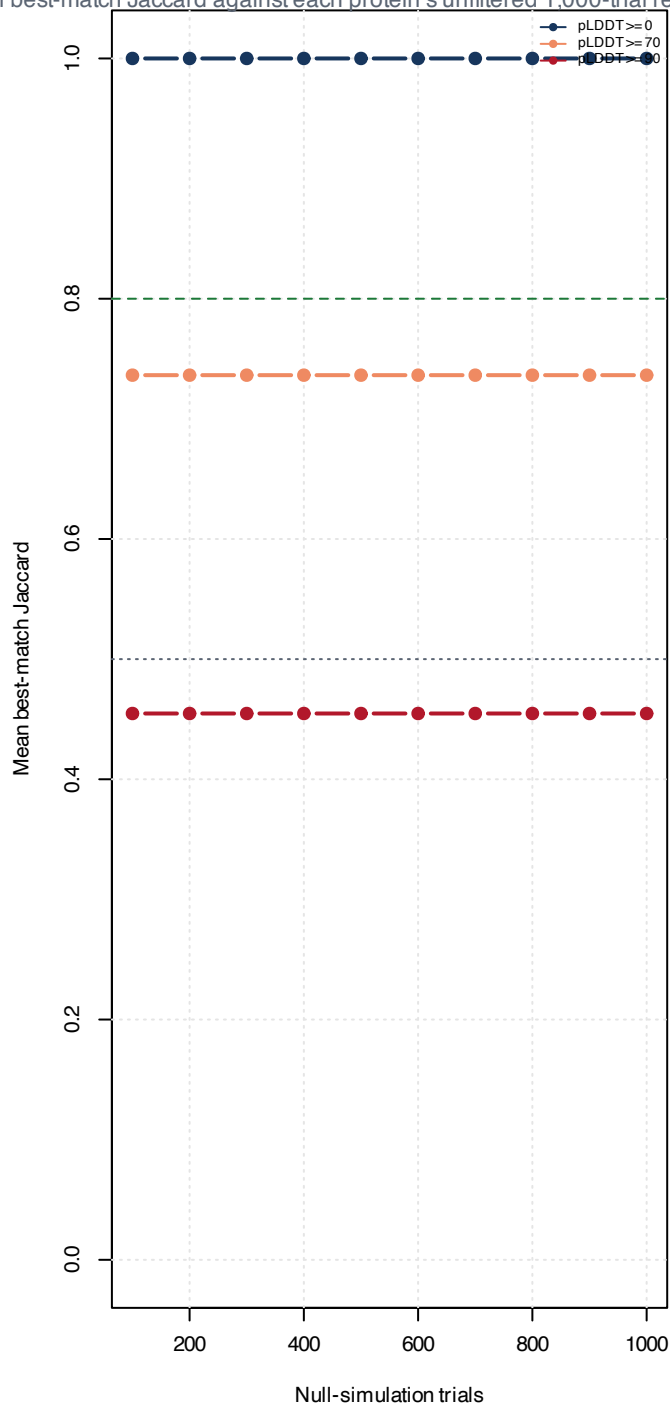
Cluster-membership stability

Mean best-match Jaccard against each protein's unfiltered 1,000-trial reference

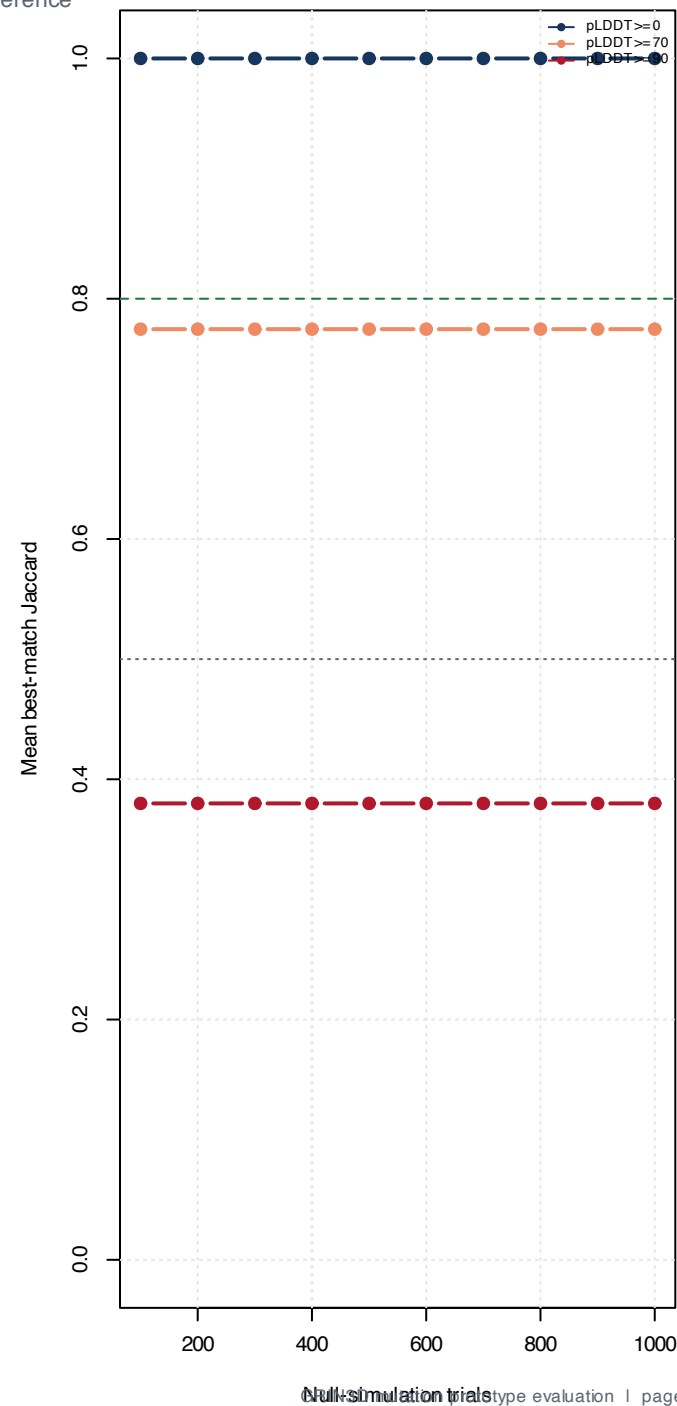
TP53



BRD4

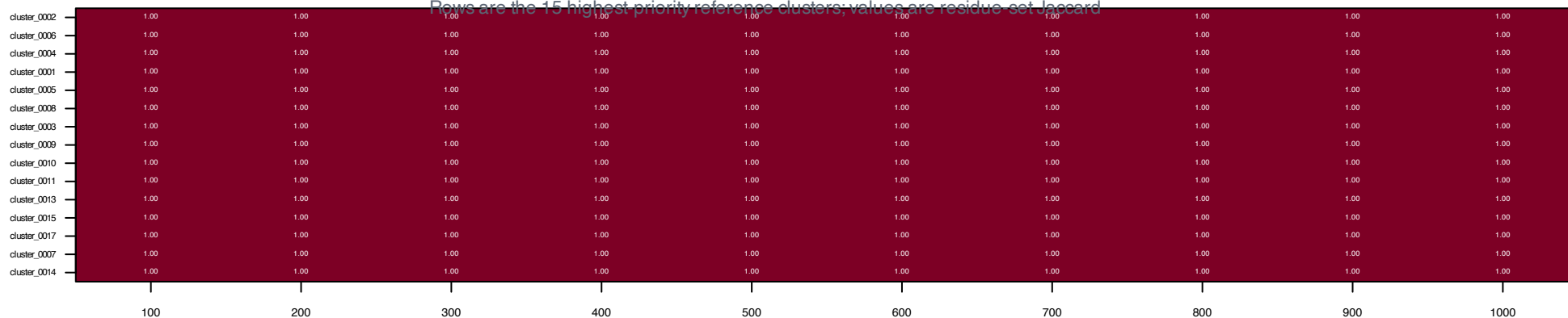


EZH2



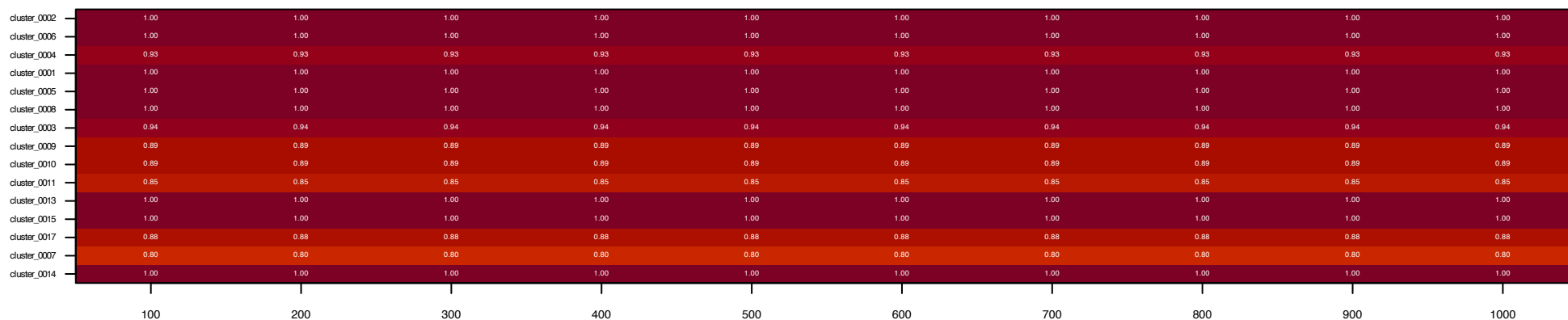
TP53 cluster-level Jaccard atlas

Rows are the 15 highest priority reference clusters; values are residue-set Jaccard



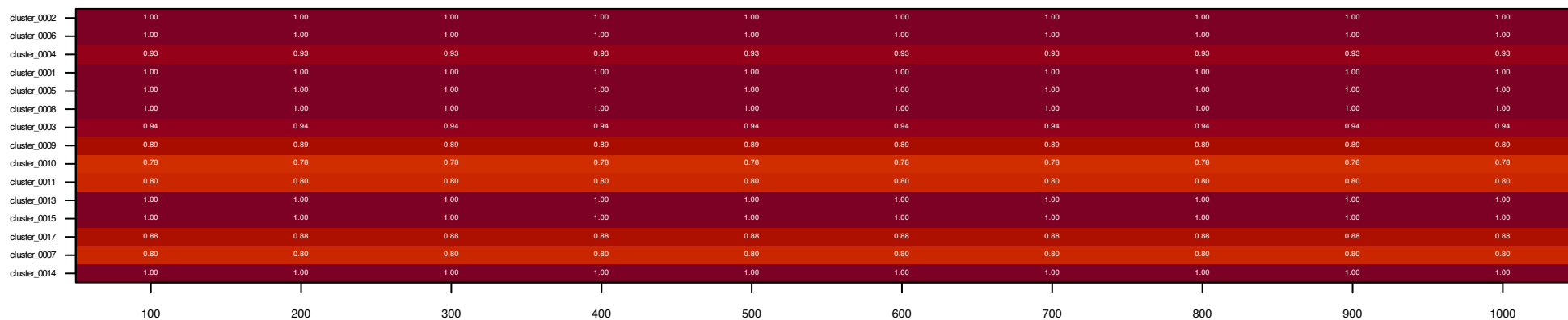
Null simulation trials

TP53 pLDDT >= 70



Null simulation trials

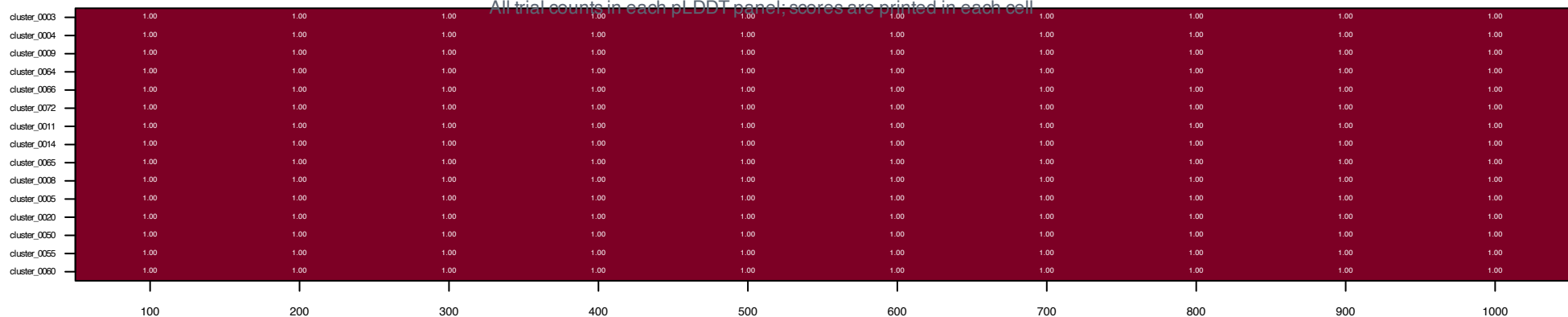
TP53 pLDDT >= 90



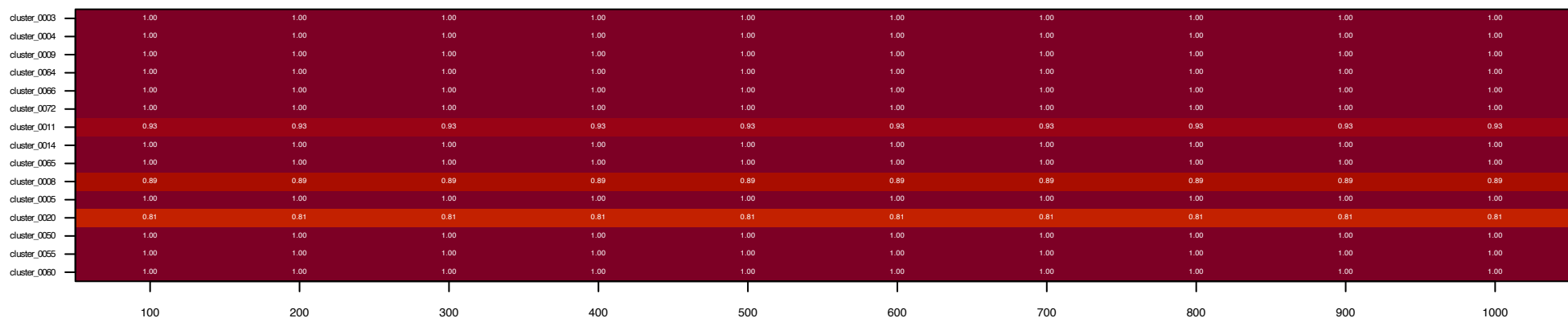
Null simulation trials

SUZ12 cluster-level Jaccard atlas

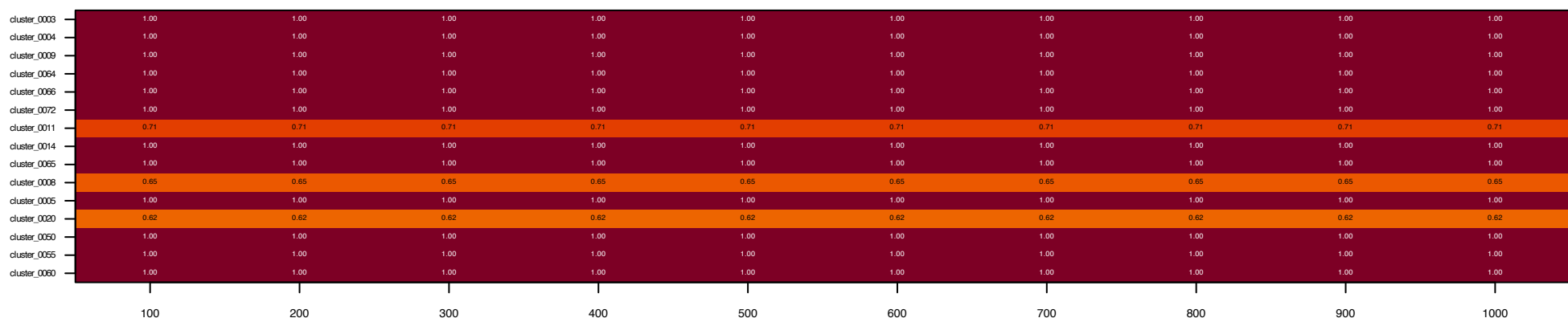
All trial counts in each pLDDT panel; scores are printed in each cell



Null-simulation trials
SUZ12 pLDDT >= 70



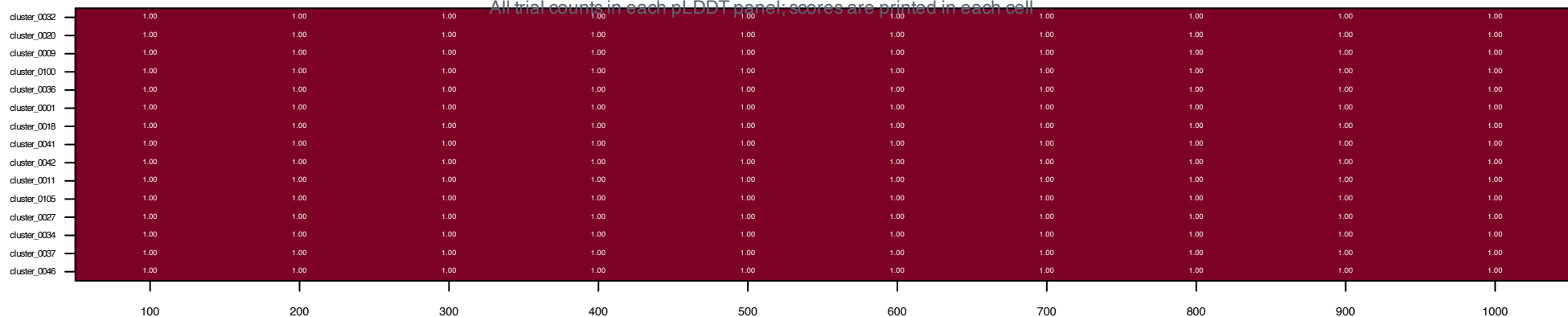
Null-simulation trials
SUZ12 pLDDT >= 90



Null-simulation trials

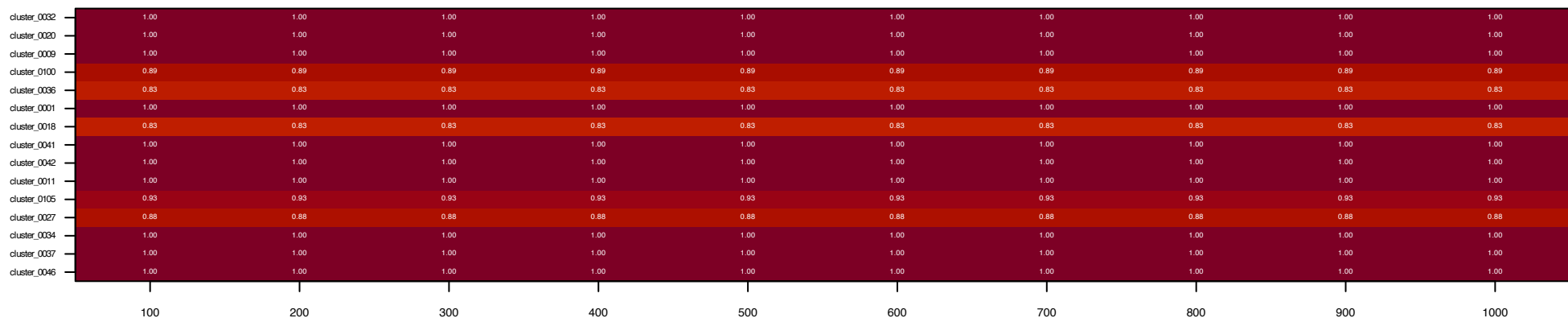
EZH2 cluster-level Jaccard atlas

All trial counts in each pLDDT panel; scores are printed in each cell



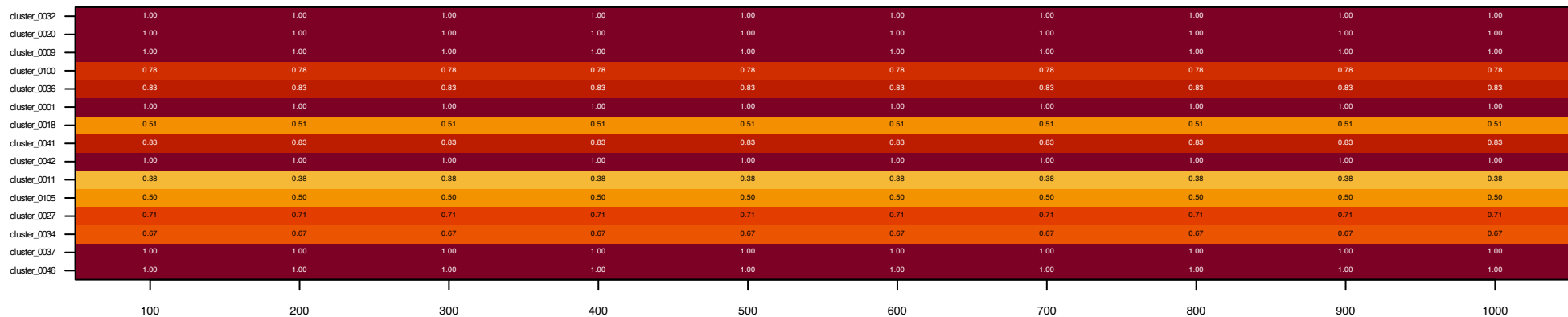
Null-simulation trials

EZH2 pLDDT >= 70



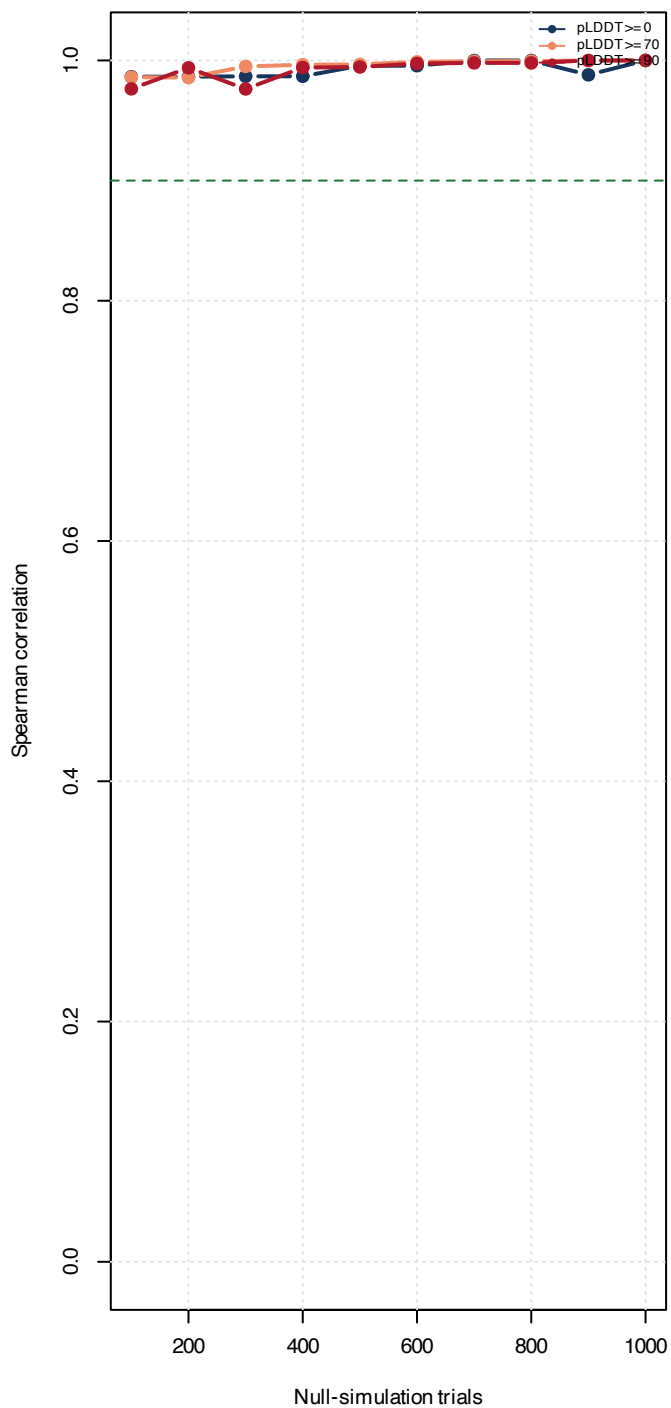
Null-simulation trials

EZH2 pLDDT >= 90



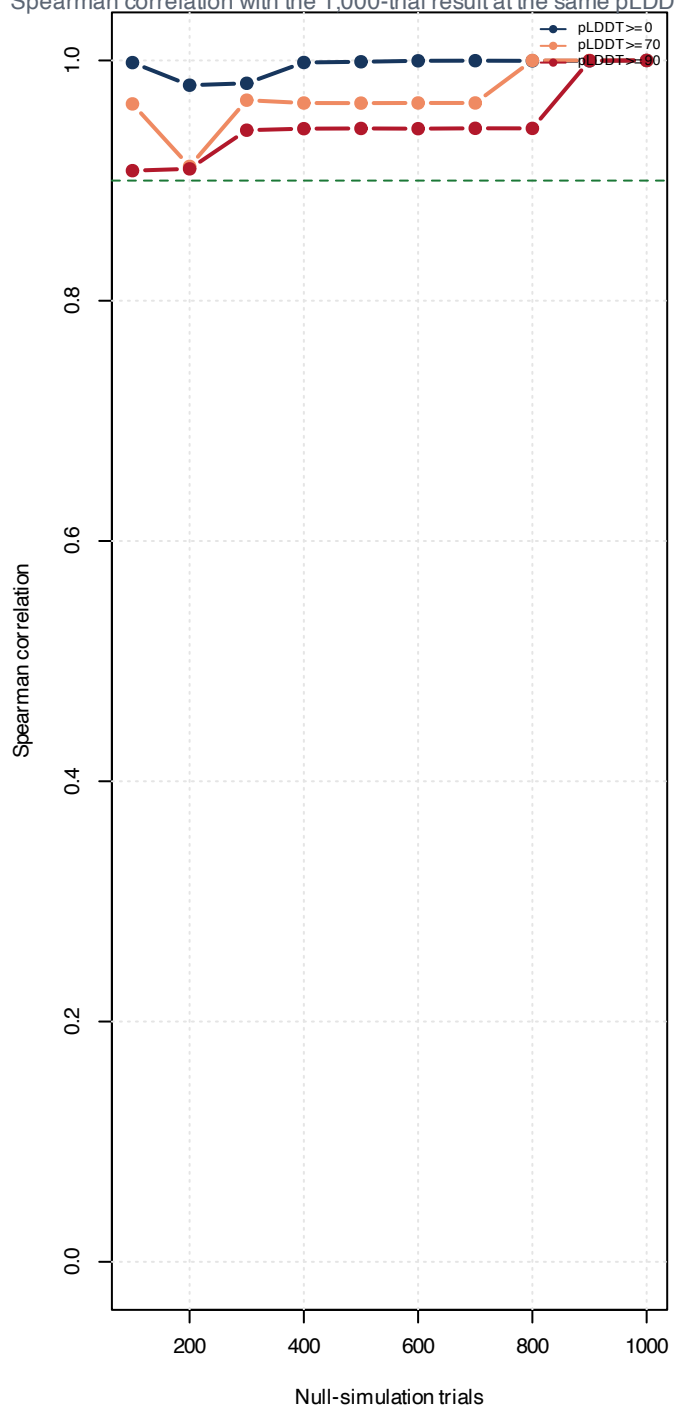
Null-simulation trials

TP53

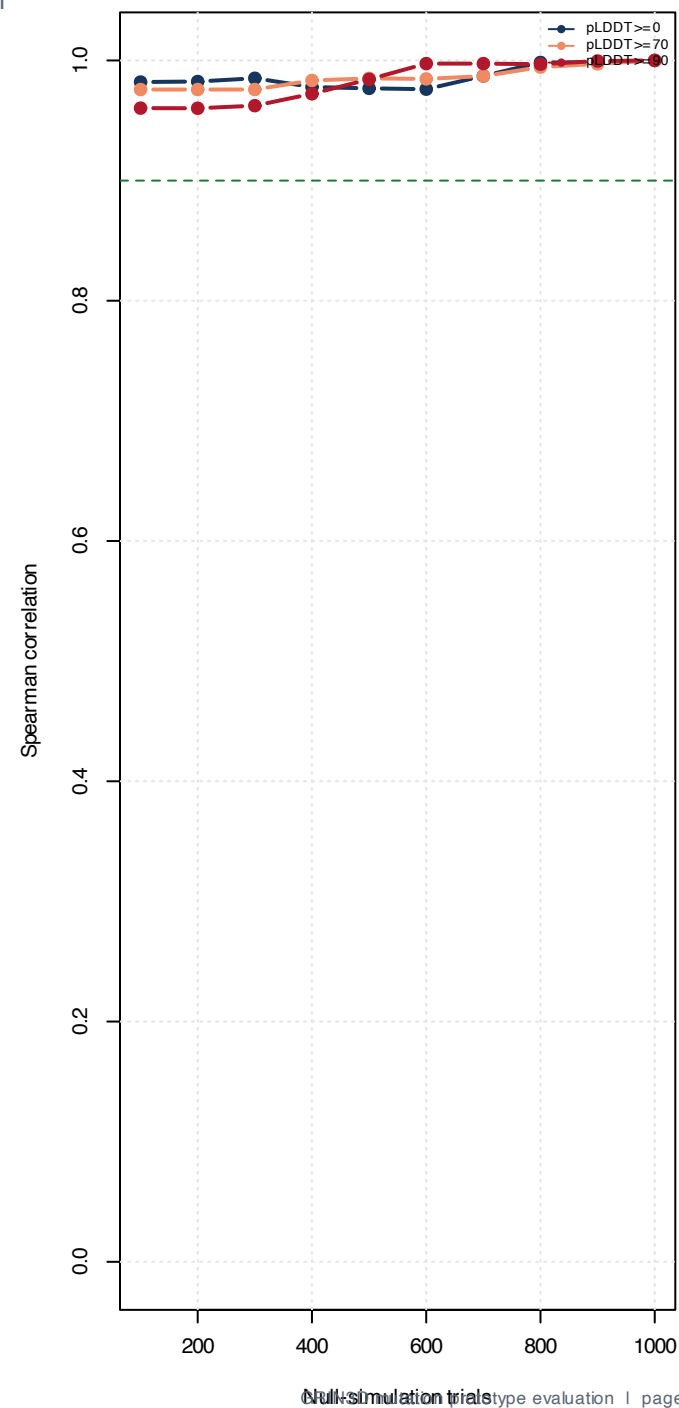


Empirical p-value rank convergence

Spearman correlation with the 1,000-trial result at the same pLDDT



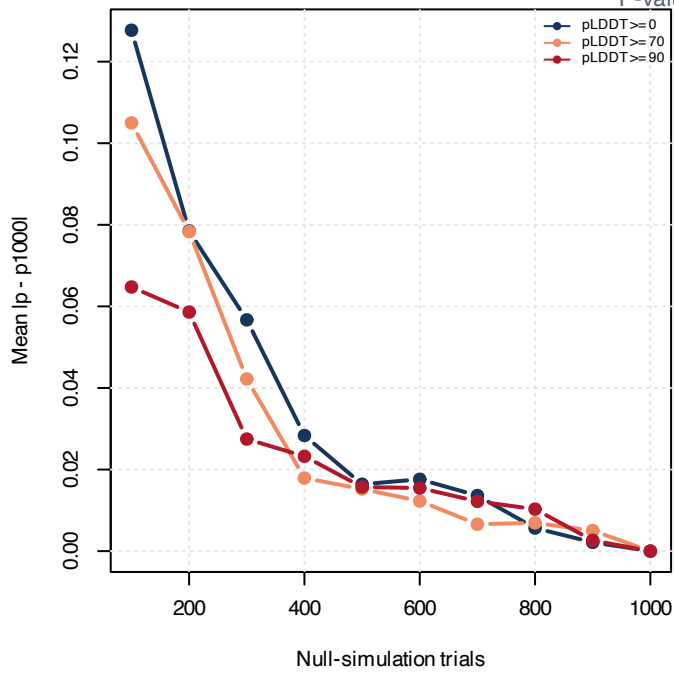
EZH2



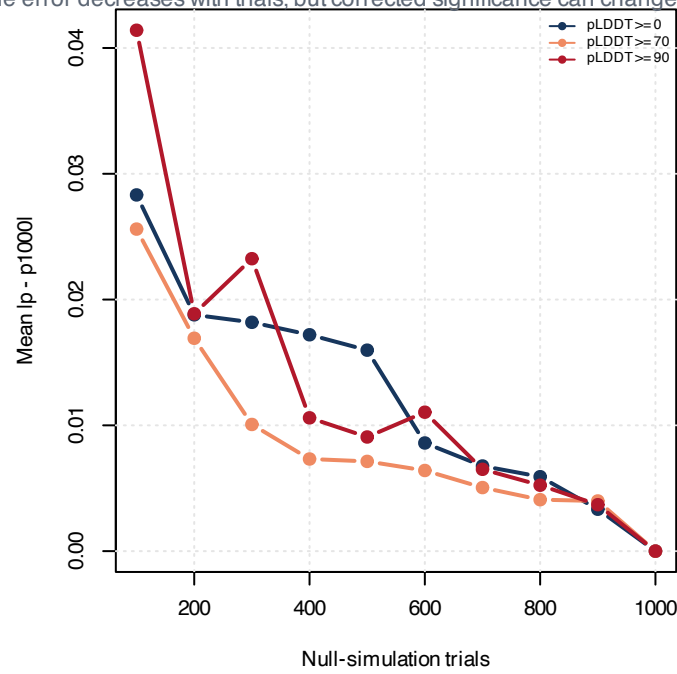
Monte Carlo convergence

P-value error decreases with trials, but corrected significance can change discretely

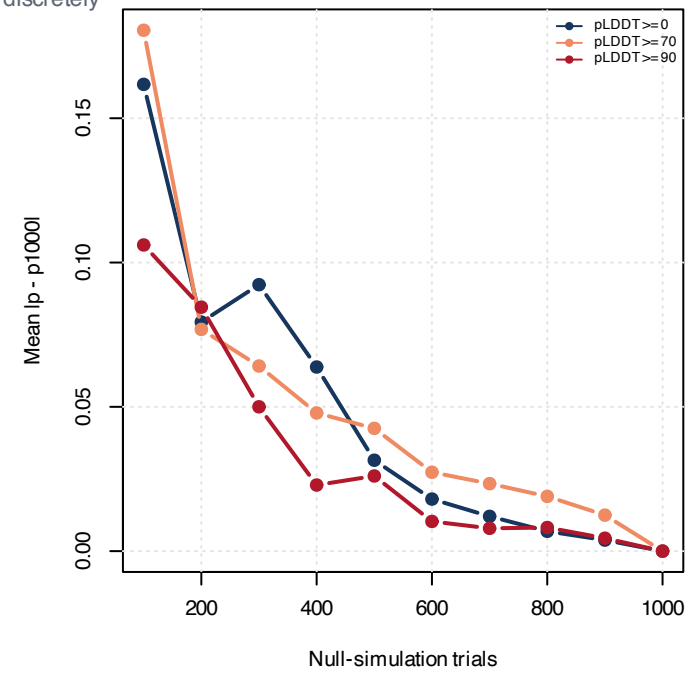
TP53 p-value error



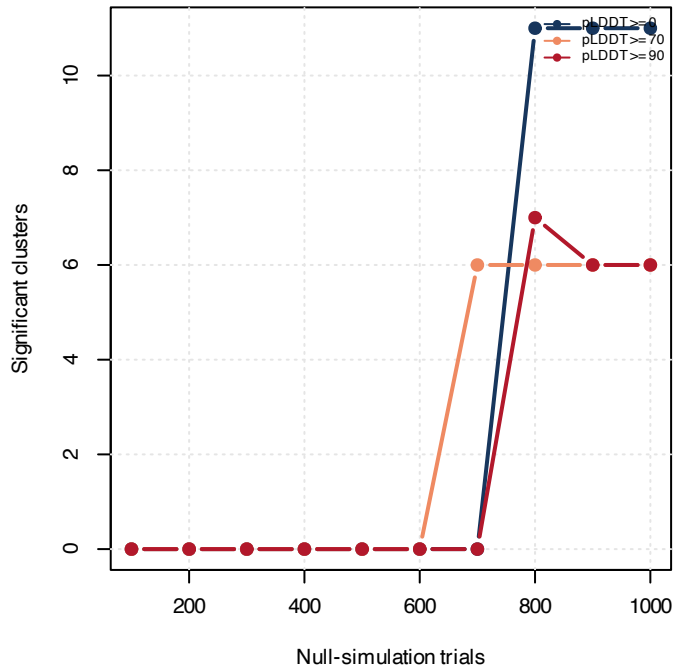
SUZ12 p-value error



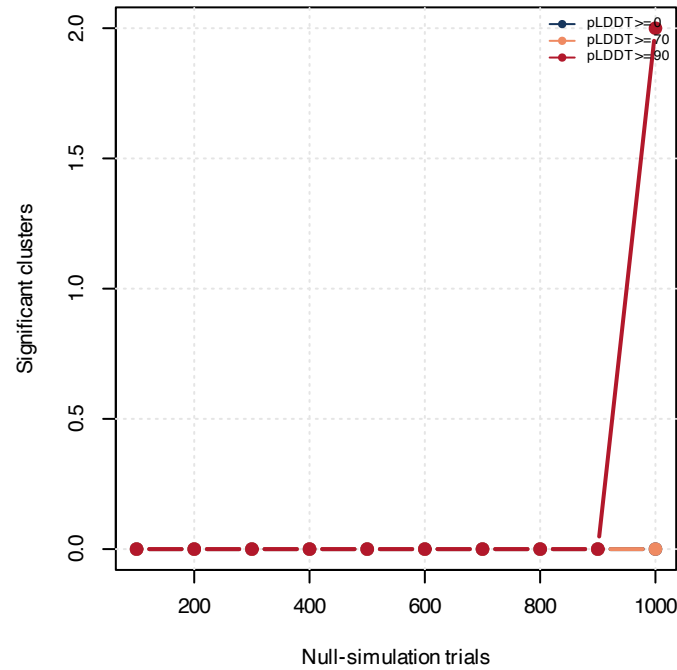
EZH2 p-value error



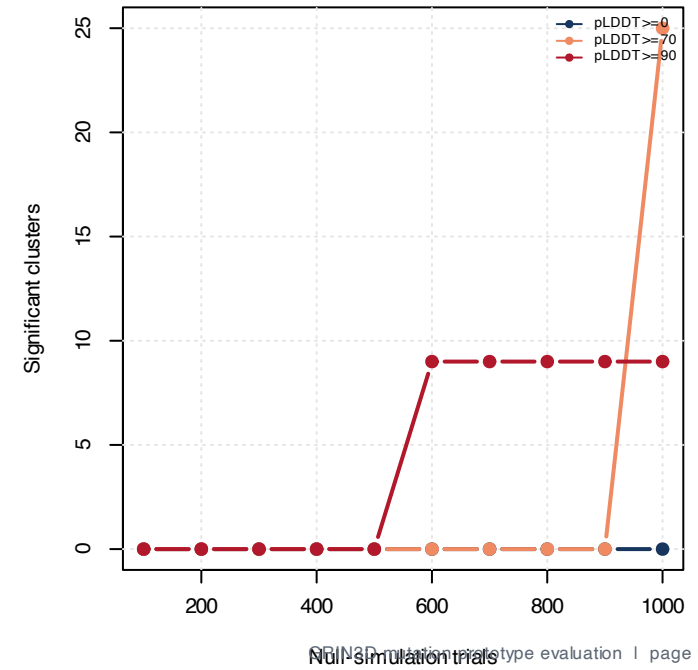
TP53 significance calls



SUZ12 significance calls



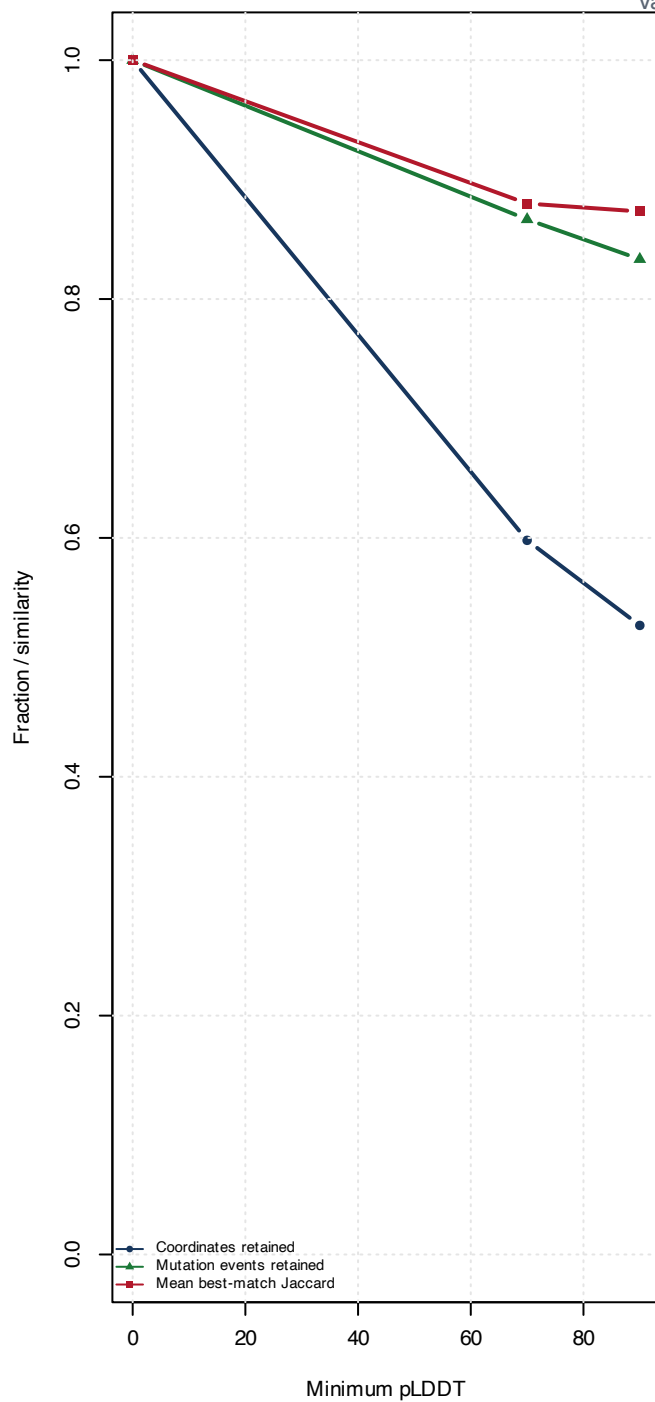
EZH2 significance calls



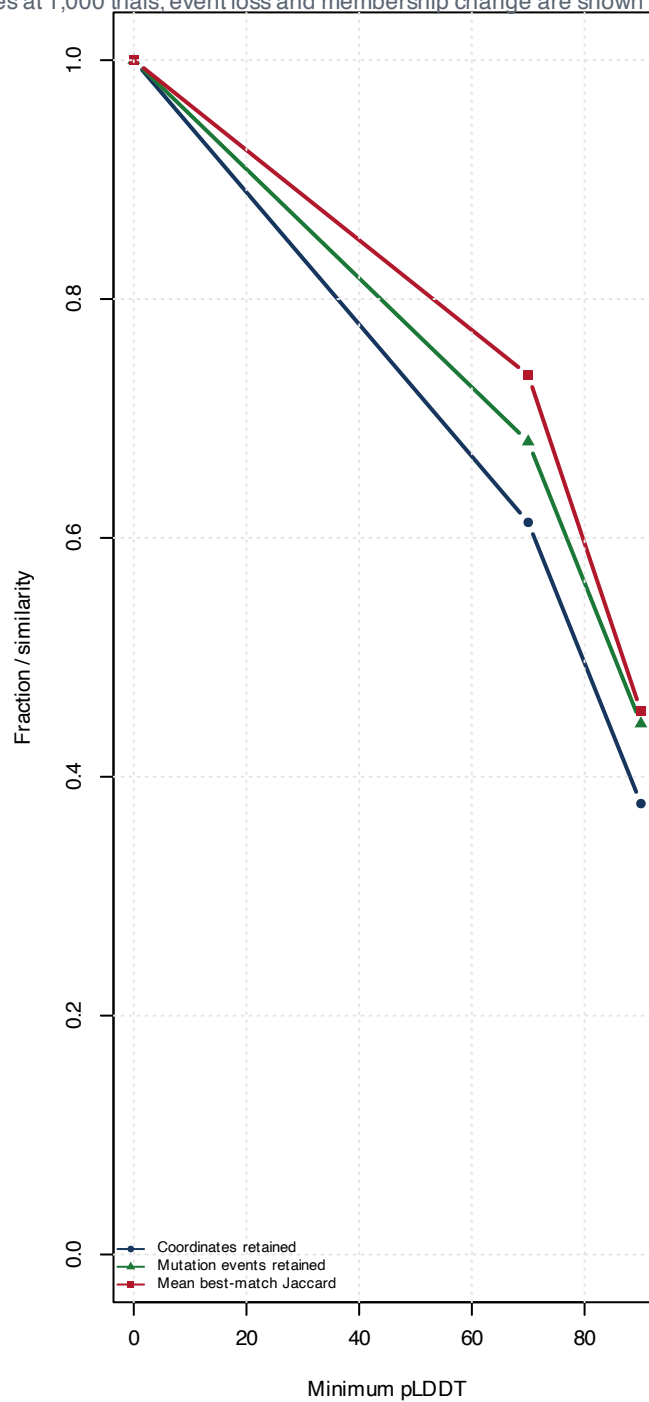
Effect of structural-confidence filtering

Values at 1,000 trials; event loss and membership change are shown together

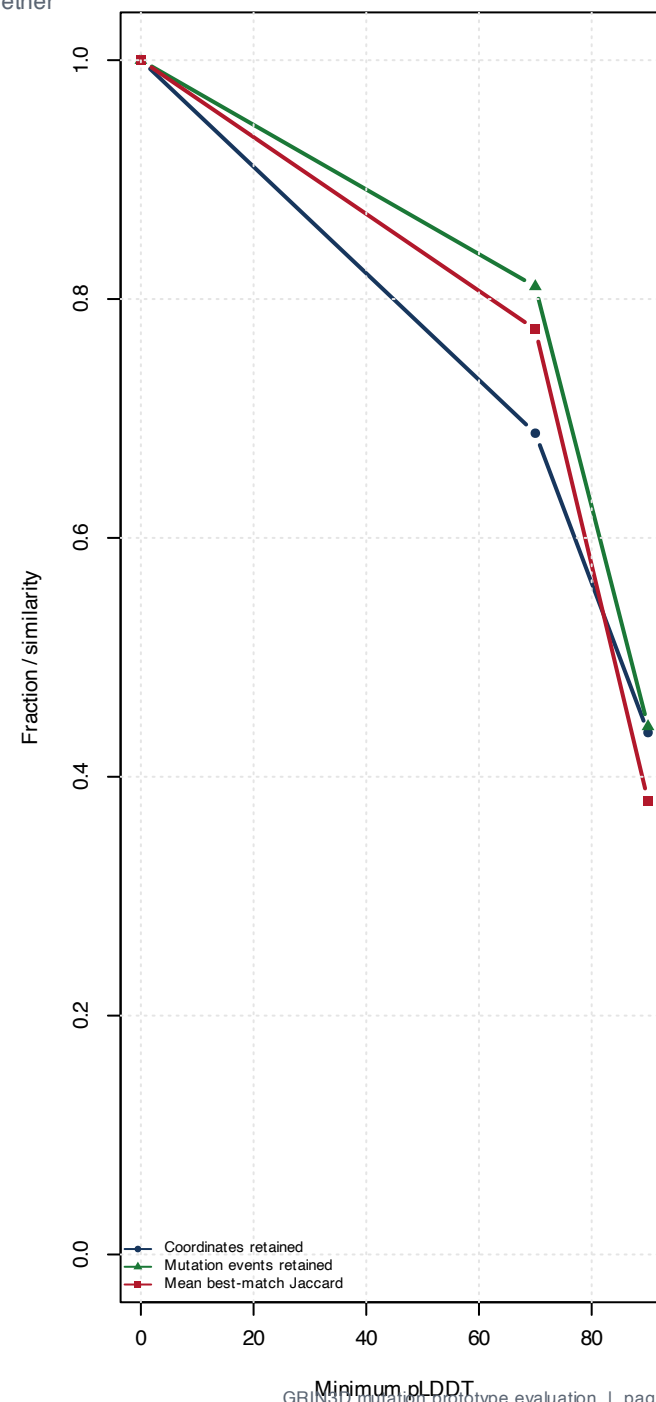
TP53



60212



EZH2



Results at 1,000 null-simulation trials

Jaccard is measured against each protein's unfiltered 1,000-trial cluster lands

Protein	pLDDT	Residues	Events	Subjects	Clusters	Significance	Mean Jaccard
TP53	0	393/393	30/30	23	41	11	1.000
TP53	70	235/393	26/30	20	35	6	0.880
TP53	90	207/393	25/30	19	33	6	0.873
SUZ12	0	739/739	72/72	57	96	0	1.000
SUZ12	70	453/739	49/72	42	69	0	0.736
SUZ12	90	279/739	32/72	26	43	2	0.455
EZH2	0	746/746	95/95	74	110	0	1.000
EZH2	70	513/746	77/95	62	82	25	0.775
EZH2	90	326/746	42/95	38	45	9	0.380

Residue and event denominators are the unfiltered input total
 changes in cluster or significance counts must be interpreted at
 filtering itself strengthens association.

When do significance calls stabilize

Earliest trial count whose exact significant-cluster set matches 1,000 trials a

Protein	pLDDT	Earliest match	Significant at 1,000
EZH2	0	100	0
EZH2	70	1000	25
EZH2	90	600	9
SUZ12	0	100	0
SUZ12	70	100	0
SUZ12	90	1000	2
TP53	0	800	11
TP53	70	700	6
TP53	90	900	6

A match first reached at exactly 1,000 trials is not evidence of convergence. A match first reached at exactly 1,000 trials is not evidence of convergence. Conditions with zero significant clusters at 1,000 trials are not evidence of convergence and should not be described as positive convergence.

Evidence synthesis

Protein	Pattern representation	Trial membership	pLDDT robustness	Statistical status
TP53	Dense positive control	Exact within pLDDT	Mean J: 0.880 / 0.873	Earliest sets: 800/700/900
SUZ12	Sparse long-protein pattern	Exact within pLDDT	Mean J: 0.736 / 0.455	Earliest sets: 100/100/1000
EZH2	Sparse long-protein pattern	Exact within pLDDT	Mean J: 0.775 / 0.380	Earliest sets: 100/1000/600

What the evidence supports

- Cluster construction is reproducible with respect to null-trial coverage.
- Increasing trial count refines empirical p-values and can change cluster residues.
- pLDDT sensitivity is biologically and statistically consequential to coverage and the retained mutation cohort.
- TP53 provides a useful positive-control pattern; SUZ12 and EZH2 provide dense hotspot example.

Defensible recommendation

Recommended use of the prototype

- Use the observed cluster tree and residue memberships as recommended, and report them independently from p-values.
- Use pLDDT ≥ 70 as a sensitivity analysis rather than silently removing coordinates, events, and subjects where pLDDT < 70 .
- Treat pLDDT ≥ 90 as a stringent stress test. It can remove coordinates, events, materially changing the estimand.
- Do not make final significance claims from the 100-1,000 trial endpoint. Instead, report trials, ideally to the prototype's final-inference target, and report the results with Monte Carlo variability.
- For each reported hotspot, show its residue set, Jaccard/contingency table, retention, empirical p-value trajectory, and biological annotations.

Bottom line: the prototype is ready for documented exploration, but it is not yet justified to label every corrected significance claim as significant. A claim first appears only at the 1,000-trial endpoint or where pLDDT ≥ 90 for all events.

Reproducibility and provenance

Analysis artifacts

- TP53 structural confidence: AlphaFold DB AF-P04637-F1 model v6 (UniProt)
- 01_run_sensitivity.R: declares and runs the trial/pLDDT grid.
- 02_summarize_visualize.R: computes Jaccard, Dice, containment, overlap
- 03_validate_analysis.R: verifies complete outputs, fixed settings, and will
- 04_analyze_trial_convergence.R: evaluates p-value and significance conve
- 05_build_pdf_report.R: builds this report from saved CSV and RDS resu

Validation

Item	Result
Configured runs	90
Result objects	90
Run manifests	90
Validation checks	25/25 passed
Trial levels	100, 200, 300, 400, 500, 600, 700, 800, 900, 1000

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annotation are intentionally maintained as separate evidence layers.